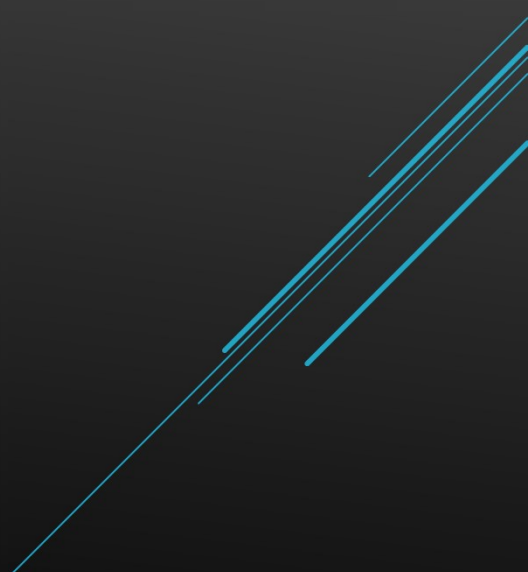


DATA SCIENCE PROJECT

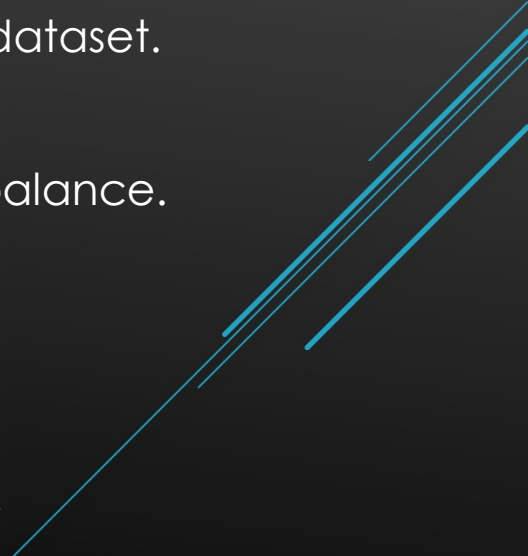
Penguins



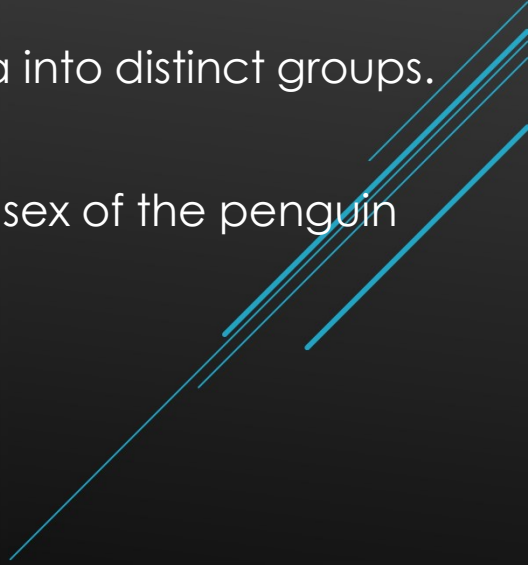
PURPOSE

- ▶ The goal of this project was to perform a full cycle of the data science process on the 'penguins' dataset.
 - ▶ Exploratory Data Analysis (EDA) was conducted, followed by model construction.
 - ▶ Insights were then gained from both the EDA phase and the model construction phase.
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EXPLORATORY DATA ANALYSIS (EDA)

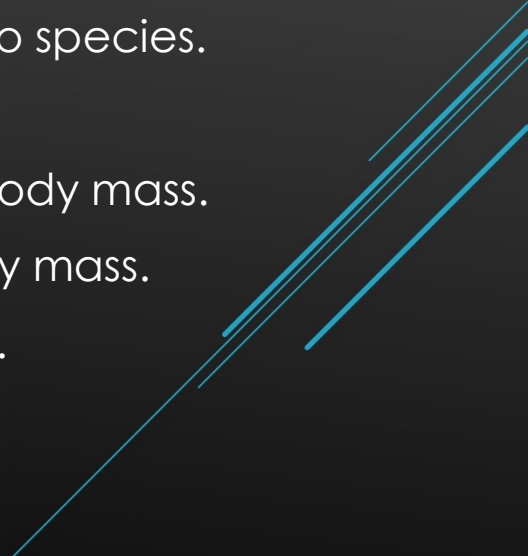
- ▶ Data discovery was first conducted to understand the structure, variables and values.
 - ▶ Data cleaning was then performed to improve the quality of the dataset.
 - ▶ Data validation assessed the dataset to ensure consistency and balance.
 - ▶ Data structured transformed the dataset into a more useful form.
 - ▶ Data visualisations provided insight on attributes and relationships.
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MODEL CONSTRUCTION AND EVALUATION

- ▶ A Simple Linear Regression model was built to predict body mass based on flipper length. The assumptions of the model were evaluated to ensure a valid model.
 - ▶ A K-Means clustering model was constructed to organise the data into distinct groups.
 - ▶ A Random Forest Classification Model was created to predict the sex of the penguin given a series of variables.
 - ▶ The models were optimized to improve performance.
 - ▶ The models were then evaluated to determine performance.
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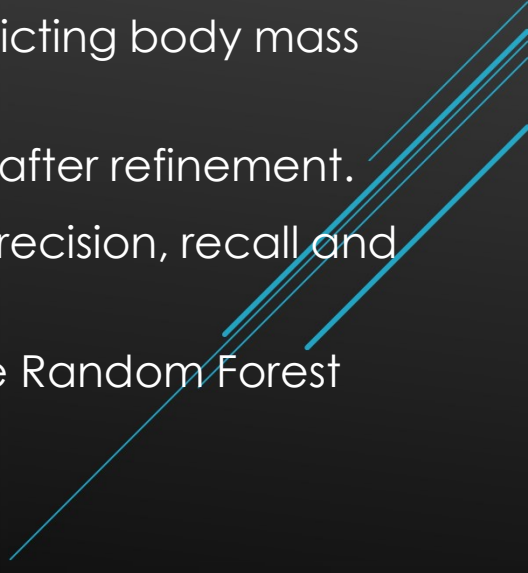
EDA INSIGHTS

The following insights were gained from the exploratory data analysis:

- ▶ The 'Gentoo' species weighs just over 1 kg more than the other two species.
 - ▶ Males are around half a kilogram heavier than females.
 - ▶ There is a strong positive correlation between flipper length and body mass.
 - ▶ Males have longer bills, longer flippers, deeper bills and more body mass.
 - ▶ Penguins with deeper bills have shorter flipper and less body mass.
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MODEL INSIGHTS

The following insights were gained from development of the models:

- ▶ The Simple Linear Regression model was reasonably good at predicting body mass given flipper length.
 - ▶ The Clustering model was not perfect but not too bad, especially after refinement.
 - ▶ The Random Forest model had a score of 92.9% over accuracy, precision, recall and F1-Score.
 - ▶ Body mass and bill depth were the most important features for the Random Forest Model, with bill length and flipper length contributing aswell.
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